HyCline, bgchm and gghybrid compared

Hybrid index and genomic cline estimates on empirical and simulated hybrid zones

Summary

- **Hybrid index: the three tools are interchangeable.** On the Italian sparrows they agree with each other at r ≥ 0.9998 and with BGC at r = 0.992; in every simulated design their errors are within 0.004 of each other. The differences are in the intervals: with low-depth reads gghybrid, which can only use called genotypes, is overconfident (87% coverage), and with 5-bird reference panels all three are (80-90%), bgchm least.
- **Genomic clines: HyCline's hierarchical model and bgchm lead, and agree with each other.** They have the smallest errors in every design (r = 0.99 between them on the sparrows), and few false outliers when the reference frequencies are well estimated. HyCline's per-locus MLE is less accurate and flags more neutral loci; gghybrid's per-locus fit has the largest errors, the lowest power for rate (v) outliers, and breaks down on centres when hybrids are skewed toward one parent.
- **The main weakness is shared: reference-panel frequencies are treated as exact.** When the panels are samples from the parental populations, as in real data, every tool flags 6-14% of neutral loci as cline outliers with 20 + 20 references, and 13-35% with 5 + 5, against 0-6% when the frequencies are exact. Hybrid indices are hardly affected.
- **Weak differentiation** leaves hybrid indices fine but makes rate outliers hard to detect; the hierarchical models then shrink toward neutral (15-18% power, almost no false positives).
- **Speed.** HyCline's whole pipeline (hybrid index, MLE and hierarchical clines) ran 3-10 times faster than bgchm or gghybrid.
- **Found and fixed along the way.** HyCline's hierarchical genomic cline fit refused genotypes missing at different individuals per locus; it now handles them (Section 5). Two benchmark settings were wrong in first attempts and were corrected (Section 6).

1. Tools and settings

Tool	Hybrid index	Genomic clines	Settings used
HyCline	maximum likelihood (estimate-hybrid-index), 95% likelihood-profile interval	per-locus MLE (fit-genomic) and hierarchical NumPyro model (fit-hierarchical-genomic)	defaults; genotype likelihoods for beagle input
bgchm (Gompert, DeRaad & Buerkle 2024)	HMC posterior (est_hi), median and 95% quantiles	hierarchical HMC model (est_genocl, hier = TRUE)	2000 iterations; est_p analytical for genotypes, HMC for genotype likelihoods
gghybrid (Bailey 2023)	MCMC posterior mode (esth), Jeffreys prior	per-locus MCMC (ggcline)	esth 3000/1000 burn-in; ggcline 5000/2000

The original bgc 1.03 could not be run (it is distributed only from its authors' site, which this environment could not reach); bgchm is its successor from the same authors. Input files for bgc 1.03 are saved alongside the others.

All three tools model a locus's allele frequency in an individual with hybrid index h as $p_0 + (p_1 - p_0)\,\phi(h)$. HyCline writes the cline as logit ϕ = u + v logit h; bgchm and gghybrid write $\phi = h^v / (h^v + (1 - h)^v e^{u'})$ with u' = -u. Clines are compared on their centre c = invlogit(-u / v) (the h at which ϕ = 0.5) and slope v.

Metrics. Hybrid index (test individuals only): mean absolute error against the true value, and the share of 95% intervals containing it. Clines: median absolute error of logit c and log v (medians, because a nearly flat cline has no identifiable centre); a locus is a v outlier when its 95% interval for v excludes 1 and a centre outlier when its interval for c excludes 0.5. Power is measured on the 20 barrier/permeable loci (v) or 10 biased loci (centre), false positives on the 70 neutral loci. HyCline's hierarchical intervals are normal approximations from its per-locus posterior mean and SD.

2. Empirical data: Italian sparrows

Hermansen et al. (2014): 77 loci, 84 house sparrow and 52 Spanish sparrow references, 433 Italian sparrows; set up as in Bailey (2023, section 2.3.1). Reference: BGC's published hybrid indices and cline parameters for the same birds and loci.

Comparison	n	Pearson r	Mean |diff|	Mean diff	Inside BGC 95% CI
HyCline vs BGC	433	0.9916	0.0287	-0.0209	0.928
gghybrid vs BGC	433	0.9916	0.0316	-0.0248	0.905
bgchm vs BGC	433	0.9917	0.0287	-0.0210	0.965
HyCline vs gghybrid	433	0.9998	0.0053	0.0039	
HyCline vs bgchm	433	0.9999	0.0017	0.0001	
gghybrid vs bgchm	433	0.9999	0.0049	-0.0038	

Comparison	Loci	log v (r)	logit c (r)	log v vs BGC beta (rho)	logit c vs BGC alpha (rho)
HyCline MLE vs BGC	77			0.819	-0.945
HyCline hierarchical vs BGC	77			0.855	-0.952
gghybrid vs BGC	77			0.775	-0.927
bgchm vs BGC	77			0.824	-0.956
HyCline MLE vs HyCline hierarchical	77	0.974	0.988		
HyCline MLE vs gghybrid	77	0.930	0.945		
HyCline MLE vs bgchm	77	0.975	0.963		
HyCline hierarchical vs gghybrid	77	0.900	0.908		
HyCline hierarchical vs bgchm	77	0.989	0.990		
gghybrid vs bgchm	77	0.902	0.847		

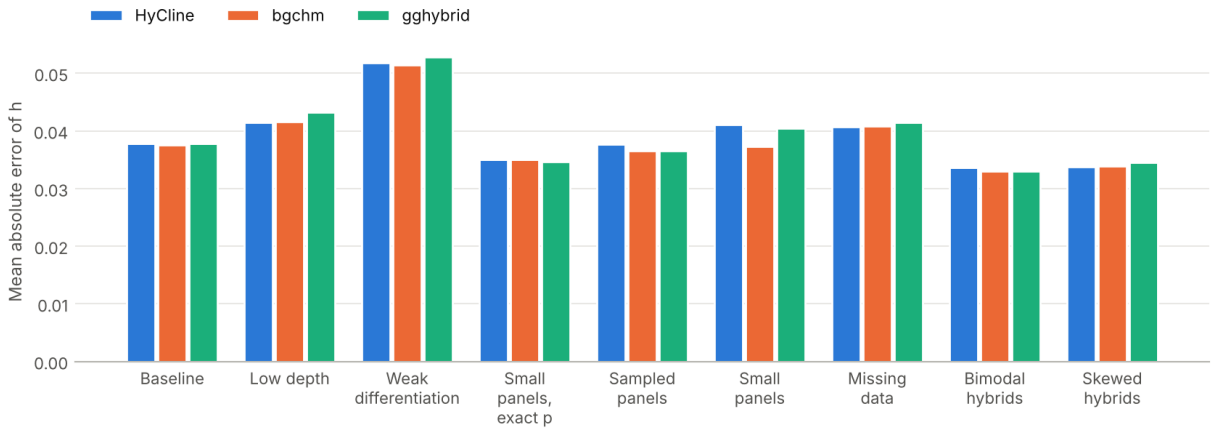
BGC's cline function differs (alpha shifts the centre, beta the rate, with opposite sign conventions), so only rank agreement is meaningful there. Run times on this data set: HyCline 195 s, gghybrid 1419 s, bgchm 2188 s (the machine was shared with other runs, so these are rough).

3. Simulated hybrid zones

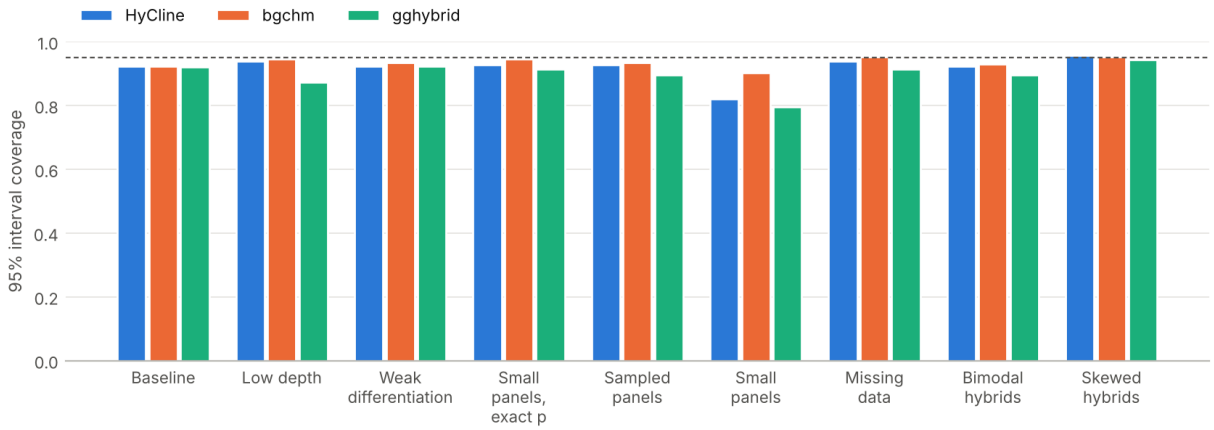
Each design changes one thing from the baseline; three replicate data sets each (seeds 1-3). Clines: 10 barrier loci (v = 3), 10 permeable (v = 0.33), 10 with a shifted centre (u = +/-1.5) and 70 neutral.

Design	What changes	Seeds
Baseline	20 + 20 parentals, 100 hybrids (uniform h), 100 loci differing by >= 0.6, true genotypes	3
Low depth	as baseline, reads at mean depth 3 (HyCline and bgchm on genotype likelihoods, gghybrid on called genotypes)	3
Weak differentiation	parental frequencies differ by >= 0.3	3
Small panels, exact p	5 + 5 parental individuals, hybrids drawn from the panels' own frequencies (the tools see exactly the frequencies the hybrids came from)	3
Sampled panels	as baseline, but panels and hybrids are drawn from population frequencies, so the 20 + 20 reference panels only estimate them (reference for Small panels)	3
Small panels	as Sampled panels, with 5 + 5 parental individuals	3
Missing data	20% of genotypes missing at random	3
Bimodal hybrids	hybrid indices ~ Beta(0.5, 0.5): mostly near the parentals	3
Skewed hybrids	hybrid indices ~ Beta(1.5, 6): mostly backcrosses toward pop1	3

3.1 Hybrid index



Mean absolute error of the test individuals' hybrid indices (lower is better), mean over seeds.

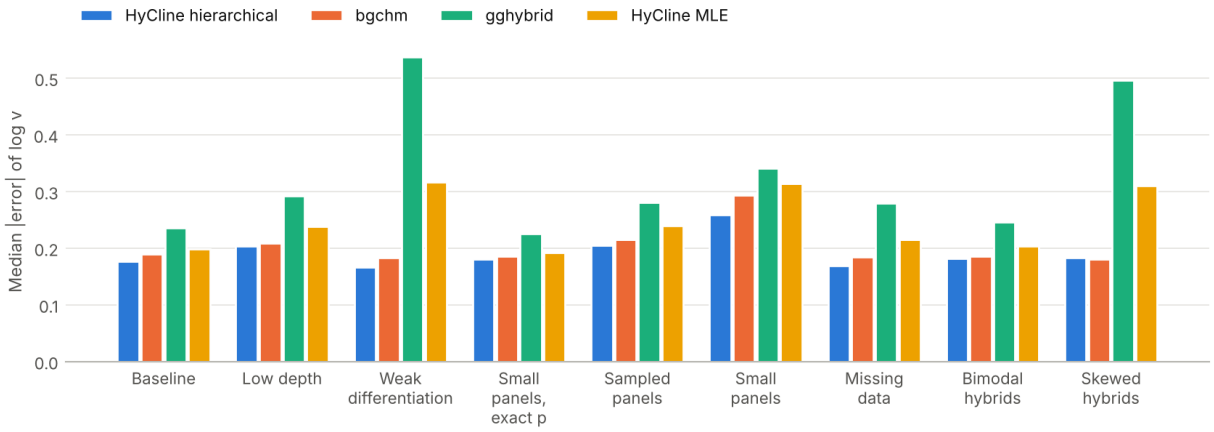


Share of 95% intervals containing the true hybrid index; dashed line at the nominal 0.95.

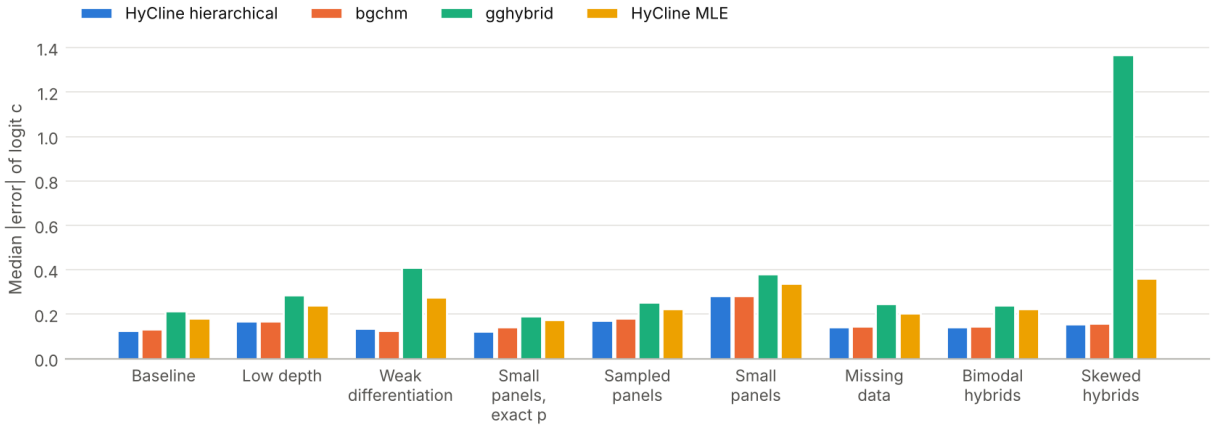
Design	Tool	Mean |error|	Bias	r	95% coverage	Interval width	Time (s)
Baseline	HyCline	0.038	0.001	0.986	0.92	0.167	91
Baseline	bgchm	0.038	0.001	0.986	0.92	0.168	279
Baseline	gghybrid	0.038	0.002	0.986	0.92	0.166	552

Design	Tool	Mean |error|	Bias	r	95% coverage	Interval width	Time (s)
Low depth	HyCline	0.041	-0.003	0.983	0.94	0.193	136
Low depth	bgchm	0.042	-0.003	0.983	0.95	0.195	862
Low depth	gghybrid	0.043	-0.003	0.982	0.87	0.172	534
Weak differentiation	HyCline	0.052	0.005	0.974	0.92	0.230	85
Weak differentiation	bgchm	0.051	0.005	0.974	0.93	0.233	566
Weak differentiation	gghybrid	0.053	0.006	0.974	0.92	0.229	781
Small panels, exact p	HyCline	0.035	-0.000	0.988	0.93	0.164	83
Small panels, exact p	bgchm	0.035	0.000	0.988	0.95	0.171	544
Small panels, exact p	gghybrid	0.035	0.000	0.988	0.91	0.163	741
Sampled panels	HyCline	0.038	0.008	0.988	0.93	0.164	71
Sampled panels	bgchm	0.037	0.008	0.988	0.93	0.165	286
Sampled panels	gghybrid	0.037	0.008	0.988	0.90	0.163	578
Small panels	HyCline	0.041	0.007	0.988	0.82	0.161	77
Small panels	bgchm	0.037	0.008	0.988	0.90	0.168	469
Small panels	gghybrid	0.040	0.008	0.988	0.80	0.159	727
Missing data	HyCline	0.041	-0.002	0.986	0.94	0.184	81
Missing data	bgchm	0.041	-0.002	0.986	0.95	0.185	411
Missing data	gghybrid	0.041	-0.002	0.986	0.91	0.184	687
Bimodal hybrids	HyCline	0.034	-0.003	0.992	0.92	0.149	83
Bimodal hybrids	bgchm	0.033	-0.003	0.992	0.93	0.150	550
Bimodal hybrids	gghybrid	0.033	-0.003	0.992	0.90	0.149	716
Skewed hybrids	HyCline	0.034	0.004	0.945	0.96	0.156	96
Skewed hybrids	bgchm	0.034	-0.001	0.945	0.95	0.157	344
Skewed hybrids	gghybrid	0.034	-0.002	0.944	0.94	0.155	468

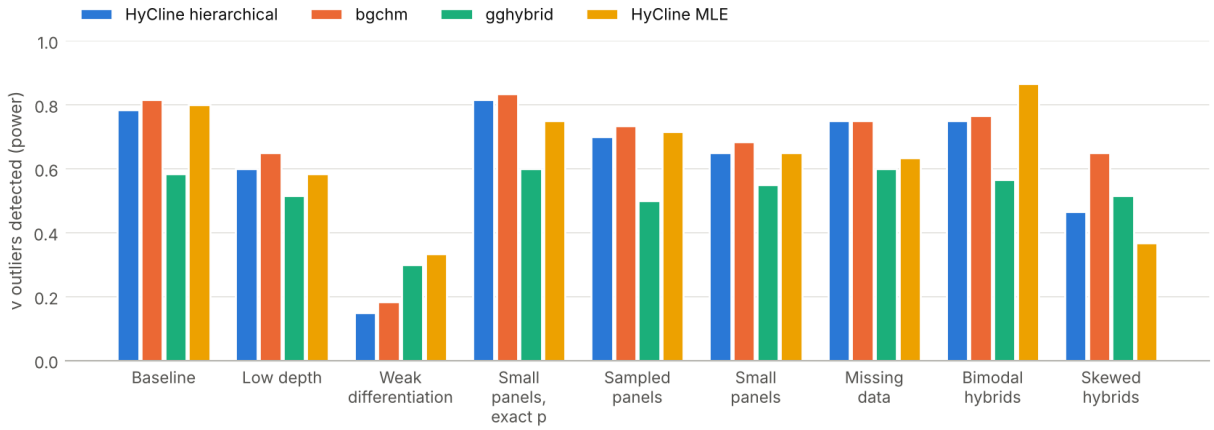
3.2 Genomic clines



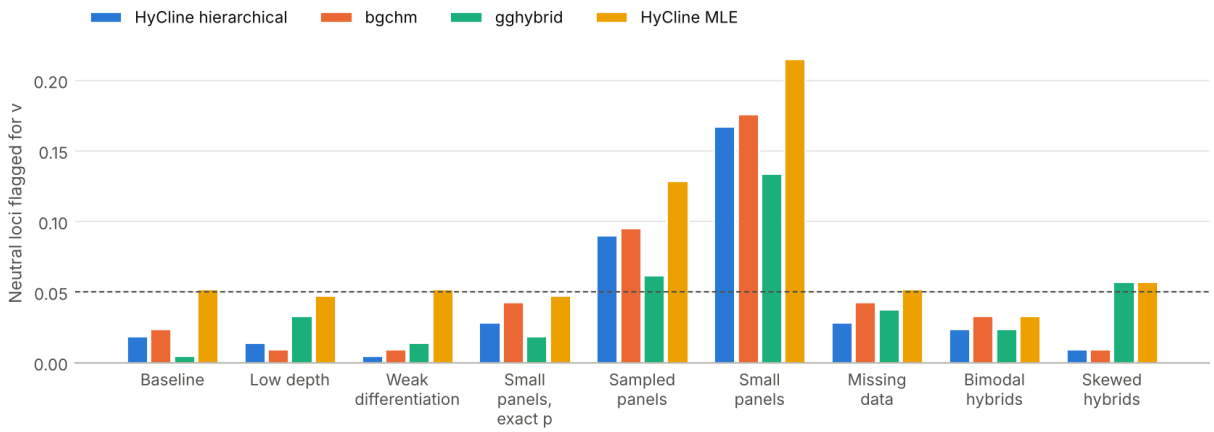
Median absolute error of log v (cline slope), mean over seeds.



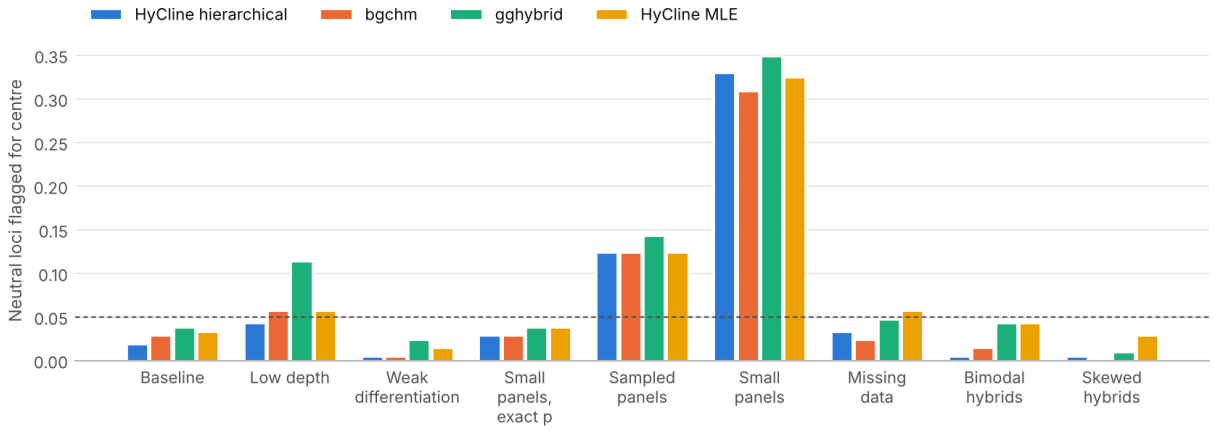
Median absolute error of logit c (cline centre).



Share of the 20 barrier and permeable loci detected as v outliers.



Share of the 70 neutral loci wrongly flagged as v outliers; dashed line at 0.05.



Share of neutral loci wrongly flagged as centre outliers.

Design	Tool	logit c |err|	log v |err|	r log v	v power	v false pos.	c power	c false pos.
Baseline	HyCline hierarchical	0.128	0.177	0.820	0.78	0.019	0.97	0.019
Baseline	bgchm	0.132	0.190	0.821	0.82	0.024	0.97	0.029
Baseline	gghybrid	0.215	0.235	0.790	0.58	0.005	0.77	0.038
Baseline	HyCline MLE	0.182	0.199	0.810	0.80	0.052	0.97	0.033
Low depth	HyCline hierarchical	0.168	0.203	0.767	0.60	0.014	0.90	0.043
Low depth	bgchm	0.167	0.209	0.760	0.65	0.010	0.90	0.057
Low depth	gghybrid	0.287	0.292	0.753	0.52	0.033	0.87	0.114
Low depth	HyCline MLE	0.240	0.238	0.718	0.58	0.048	0.87	0.057
Weak differentiation	HyCline hierarchical	0.135	0.166	0.660	0.15	0.005	0.43	0.005
Weak differentiation	bgchm	0.126	0.183	0.666	0.18	0.010	0.37	0.005
Weak differentiation	gghybrid	0.412	0.537	0.575	0.30	0.014	0.30	0.024
Weak differentiation	HyCline MLE	0.275	0.317	0.548	0.33	0.052	0.50	0.014
Small panels, exact p	HyCline hierarchical	0.122	0.180	0.830	0.82	0.029	1.00	0.029
Small panels, exact p	bgchm	0.143	0.185	0.823	0.83	0.043	0.97	0.029
Small panels, exact p	gghybrid	0.191	0.225	0.775	0.60	0.019	0.83	0.038

Design	Tool	logit c |err|	log v |err|	r log v	v power	v false pos.	c power	c false pos.
Small panels, exact p	HyCline MLE	0.176	0.192	0.690	0.75	0.048	0.87	0.038
Sampled panels	HyCline hierarchical	0.173	0.205	0.770	0.70	0.090	1.00	0.124
Sampled panels	bgchm	0.181	0.216	0.765	0.73	0.095	1.00	0.124
Sampled panels	gghybrid	0.254	0.280	0.766	0.50	0.062	0.90	0.143
Sampled panels	HyCline MLE	0.225	0.239	0.735	0.72	0.129	0.97	0.124
Small panels	HyCline hierarchical	0.283	0.259	0.665	0.65	0.167	0.87	0.330
Small panels	bgchm	0.281	0.293	0.629	0.68	0.176	0.83	0.310
Small panels	gghybrid	0.380	0.340	0.570	0.55	0.134	0.80	0.349
Small panels	HyCline MLE	0.337	0.313	0.544	0.65	0.215	0.77	0.325
Missing data	HyCline hierarchical	0.143	0.168	0.824	0.75	0.029	1.00	0.033
Missing data	bgchm	0.145	0.184	0.820	0.75	0.043	1.00	0.024
Missing data	gghybrid	0.249	0.279	0.828	0.60	0.038	0.87	0.048
Missing data	HyCline MLE	0.206	0.214	0.725	0.63	0.052	0.93	0.057
Bimodal hybrids	HyCline hierarchical	0.142	0.182	0.810	0.75	0.024	0.90	0.005
Bimodal hybrids	bgchm	0.146	0.186	0.828	0.77	0.033	0.83	0.014
Bimodal hybrids	gghybrid	0.242	0.246	0.774	0.57	0.024	0.77	0.043
Bimodal hybrids	HyCline MLE	0.224	0.203	0.774	0.87	0.033	0.83	0.043
Skewed hybrids	HyCline hierarchical	0.155	0.183	0.770	0.47	0.010	0.63	0.005
Skewed hybrids	bgchm	0.159	0.180	0.789	0.65	0.010	0.63	0.000
Skewed hybrids	gghybrid	1.368	0.497	0.723	0.52	0.057	0.27	0.010
Skewed hybrids	HyCline MLE	0.360	0.311	0.666	0.37	0.057	0.70	0.029

3.3 What each design shows

- **Low depth (3x).** Hybrid index barely changes for HyCline and bgchm, which use genotype likelihoods (error 0.041-0.042, coverage 0.94-0.95); gghybrid on called genotypes has similar error but 87% coverage. Cline power drops for every tool (v outliers 52-65%).
- **Weak differentiation (≥ 0.3).** Hybrid-index error rises to about 0.052 for all three. Rate outliers become hard to detect: the hierarchical models shrink toward neutral (power 0.15-0.18, false positives ≤ 0.01); HyCline's MLE finds more (0.33) with 5% false positives; gghybrid's errors roughly double.
- **Sampled panels (20 + 20) and small panels (5 + 5).** With panels drawn from the parental populations, cline false-positive rates rise for every tool (6-14% with 20 per panel, 13-35% with 5), and hybrid-index coverage falls with 5 per panel (HyCline 0.82, gghybrid 0.80, bgchm 0.90). With exact panel frequencies, 5 per panel costs nothing (Small panels, exact p), so the loss comes entirely from estimating the parental frequencies.
- **Missing data (20%).** Little effect on any tool; all handle it (HyCline's hierarchical model after the fix in Section 5).
- **Bimodal hybrids.** Hybrid indices are slightly more accurate (most birds are near 0 or 1); clines are similar to the baseline.
- **Skewed hybrids.** Hybrid indices stay accurate. With few birds above h = 0.5, centres are poorly informed: the hierarchical models stay accurate (logit-centre error 0.155-0.159) while HyCline's MLE (0.36) and especially gghybrid (1.37) do not; bgchm detects more rate outliers (0.65) than HyCline's hierarchical model (0.47).

5. HyCline limitation found and fixed

With 20% of genotypes missing, *fit-hierarchical-genomic* stopped with an error: it required every locus to be observed at the same individuals, while bgchm and gghybrid fit the data. Loci are now aligned by individual and a missing genotype enters the likelihood with zero sampled alleles, which leaves it out exactly. A regression test built from the simulated data covers it; the missing-data results in this report use the fixed version.

6. Corrections and caveats

- **bgchm setting, corrected.** The first low-depth run estimated bgchm's parental allele frequencies from genotype likelihoods with its analytical shortcut (est_p, HMC = FALSE). That shrank them toward 0.5 (mean |p - 0.5| 0.29 against 0.36 in the panels) and inflated bgchm's hybrid-index error to 0.064. The runs reported here use HMC, bgchm's default for genotype likelihoods.
- **Simulator setting, corrected.** The first small-panel run drew hybrids from the reference panels' own frequencies, so the tools were given exactly the frequencies the hybrids came from and panel size had no effect. The small-panel and sampled-panel designs draw panels and hybrids from population frequencies instead; the first run is kept as Small panels, exact p.
- **gghybrid at low depth** takes genotypes only, so it was given genotypes called from the reads; its narrower intervals there reflect treating those calls as certain.
- **HyCline's hierarchical intervals** are normal approximations from the per-locus posterior mean and SD, which is all its output gives; bgchm's and gghybrid's are posterior quantiles.
- **Run times** are wall-clock on 4 cores for each tool's whole pipeline. Several runs shared the machine, so compare times within a design, not across designs.
- **Scope.** 100 loci and 100 hybrids per data set, three seeds per design; small differences (a few thousandths in hybrid-index error, a few percentage points in power) are within replicate variation.

7. Files

Everything needed to rerun or rescore is in *benchmarking/tool_comparison/*: one folder per design with, for each data set, the simulation truth (*sim/*), the inputs and run scripts for HyCline, gghybrid, bgchm and bgc 1.03 (*inputs/*), the scored outputs and the timings. *python benchmarking/compare_tools.py --score-only --reps 3 --outdir benchmarking/tool_comparison/<design>* reproduces the scores; *benchmarking/italy_sparrow_tools.py* reruns the empirical comparison from a gghybrid checkout (its scores are in *tool_comparison/italy_sparrows/*; the sparrow genotypes themselves are not copied into HyCline, which is MIT-licensed, from gghybrid's GPL-3 repository).